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Rock magnetic evidence for inclination shallowing in the early Carboniferous Deer Lake Group red beds of western Newfoundland

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SUMMARY

A paleomagnetic and rock magnetic study of the Carboniferous Deer Lake Group red beds of Newfoundland was performed to detect and correct for inclination shallowing. Results indicate a primary remanence carried by magnetite, with a mean direction of $D = 179.7^\circ$, $I = 33.7^\circ$, $\alpha_{95} = 7.2^\circ$ which corresponds to a paleopole position of 22.2°N, 122.3°E, $A_{95} = 7.6^\circ$. Correcting the inclination using anisotropy of anhysteretic remanence and the measured individual particle anisotropy gives a corrected direction of $D = 178.8^\circ$, $I = 50.9^\circ$, $\alpha_{95} = 6.3^\circ$ corresponding to a paleopole position at 8.4°N, 122.7°E, $A_{95} = 7.2^\circ$. This correction is larger than that of other red beds from the Maritime Provinces of Canada, but is consistent with paleoenvironmental reconstructions, placing North America in a more arid climate zone. Our inclination-corrected results have important implications for this portion of North America's apparent polar wander path and suggest a correction is needed for other red bed-derived APWPs. We have determined the range of flattening factors f, defined as the proportionality constant between the tangents of the measured (I_m) and field (I_o) inclinations, $\tan(I_m) = f\tan(I_0)$, from this study and previous inclination correction studies to estimate inclination corrections. Using the range of haematite f factors observed in this study to correct the Neogene red bed inclinations from the Vallès-Penedès Basin (NE Spain) yields inclinations consistent with the known geomagnetic field inclination in the Neogene, thus indicating that the range of f factors reported here may be used to estimate the magnitude of inclination shallowing in red beds.

Key words: Magnetic fabrics and anisotropy; Magnetic mineralogy and petrology; Palaeomagnetism applied to tectonics; Palaeomagnetism applied to geologic processes; Rock and mineral magnetism.

1 INTRODUCTION

The stable remanence of red beds has long been studied by paleomagnetists because of red beds' frequent occurrence in continental sedimentary records. The apparent polar wander paths (APWPs) of some continents, for instance, such as the Meso-Paleozoic portion of North America's APWP, are dominated by red bed-derived paleopoles (Van der Voo 1990; McElhinny & Lock 1993, 1996).

The mechanism by which red beds acquire their remanence, and therefore the accuracy with which red beds record paleomagnetic directions, is poorly understood (Collinson 1965; Dunlop & Özdemir 1997). The most common view is that the remanence is carried by haematite of chemical origin, in which case the remanence will be a chemical remanent magnetization (CRM) that was acquired either soon after or up to millions of years after deposition (Collinson 1974; Dunlop & Özdemir 1997). The timing of the acquisition of the remanence is a fundamental parameter in the interpretation of a paleomagnetic study (Roy & Park 1972; Walker *et al.* 1981; Larson *et al.* 1982; Liebes & Shive 1982).

In some cases, the remanence is carried by detrital haematite, making the remanence a primary detrital remanent magnetization (DRM) (Collinson 1974; Elston & Purucker 1979; Steiner 1983; Tauxe *et al.* 1990; Tan & Kodama 2002; Tan *et al.* 2007; Bilardello 2008; Bilardello & Kodama 2009). In many red beds, however, different magnetic mineralogies often coexist and different combinations of primary and secondary mineralogies can occur, making paleomagnetic studies of red beds case specific. For instance, pigmentary haematite is typically present in red beds, giving them their characteristic red colouration, and is secondary in origin (Dunlop & Özdemir 1997; McElhinny & McFadden 1999; Kodama & Dekkers 2004). Other red beds may have a primary remanence carried by magnetite or maghemite, with pigmentary haematite (and other mineralogies) present as secondary phases (see Dunlop & Özdemir 1997). For any of the earlier cases, the primary remanence may be affected by depositional and/or post-depositional processes. Burial compaction, which can shallow the inclination of the magnetization, is the focus of this paper (Anson & Kodama 1987; Sun

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& Kodama 1992). Shallowing of the paleomagnetic inclination is well-documented in red beds (Garcés *et al.* 1996; Gilder *et al.* 2001; Tan & Kodama 2002, 2003; Tauxe & Kent 2004; Kent & Tauxe 2005; Vaughn *et al.* 2005; Tan *et al.* 2007). Inclinations that are too shallow result in erroneous paleolatitude estimates and strongly affect the APWPs of continents, hence the need to correct for this error (Garcés *et al.* 1996; Gilder *et al.* 2001; Tan & Kodama 2003; Vaughn *et al.* 2005; Tan *et al.* 2007).

Two inclination shallowing correction techniques have been proposed. One is based on magnetic anisotropy measurements (Jackson *et al.* 1991; Kodama 1997; Tan & Kodama 2002, 2003; Vaughn *et al.* 2005; Tan *et al.* 2007), the other is based on models of the geomagnetic field secular variation [the elongation–inclination (EI) technique] (Tauxe & Kent 2004; Kent & Tauxe 2005). Anisotropy-based corrections rely on the assumption that the orientation distribution of magnetic particle easy axes in sedimentary rocks is nearly isotropic soon after deposition and acquires a bedding parallel foliated anisotropy during burial compaction (Stephenson *et al.* 1986; Jackson *et al.* 1991; Tan & Kodama 2003). The magnetic fabric, together with the individual particle anisotropy is used to correct the inclination (Jackson *et al.* 1991; Tan & Kodama 2002, 2003; Tan *et al.* 2003). Corrections that are based on models of the geomagnetic field, on the other hand, rely on the expected distribution of paleomagnetic directions caused by secular variation. During a period long enough to observe secular variation ($\sim 10^3$ years), paleomagnetic directions will be elongated in a N–S direction, with the amount of elongation depending on the sampling site's initial paleolatitude, with the maximum elongation occurring at the equator. Shallowed inclinations, however, will result in flattened distributions that are elongate in an E–W direction. Tauxe & Kent (2004) have developed a correction technique that relies on models of the geomagnetic field to restore the elongation and inclination of the distribution.

In this paper, we perform a paleomagnetic and rock magnetic study of the early Carboniferous Deer Lake Group red beds from Newfoundland to detect and correct for inclination shallowing. Results are compared to other Carboniferous red bed-derived paleopoles from the Maritime Provinces of Canada that have been corrected for inclination shallowing (Bilardello 2008; Bilardello & Kodama 2010). The effects of the corrections on the Carboniferous APWP for North America are evaluated in terms of the paleoenvironmental reconstructions of the rocks sampled. Finally, we determine the relationship between corrected inclinations and the amount of shallowing for all available inclination-corrected results. We check the validity of this relationship by applying a simplified correction to the Neogene Vallès-Penedès Basin (NE Spain) red beds.

2 GEOLOGY

The Early Carboniferous Deer Lake Group is part of the fresh water sedimentary rocks that fill the Deer Lake Basin and rest on Precambrian (Grenville Structural Province of the Canadian Shield) basement gneisses. The deposits of the Deer Lake Basin have been subdivided by Hyde (1984) into the Tournaisian Anguille Group, the Wigwam Brook and Wetstone Point Formations, the unconformably overlying predominantly Visean Deer Lake Group and the Westphalian Howley Group (Fig. 1).

The four units of the Deer Lake Group are well exposed and are best suited for a paleomagnetic study. All four units were sampled for this study. The lowermost unit, the North Brook Formation is an over 2000-m-thick fluvial and alluvial fan deposit (Belt 1969; Hyde 1979), composed of mainly red to grey arkosic sandstone and conglomerate, with minor red to grey siltstone and pink to grey micritic limestone. The Rocky Brook Formation lies conformably on top of the North Brook, and its lowermost layers intertongue with facies equivalent to the North Brook Formation. It is typically 500- to 600-m thick and consists of red calcareous silts, grey to green siltstones and mudstones, calcareous dolostones, dolomitic limestones, and grey, green and black mudstones. The unit is interpreted by Hyde (1984) to be mixed lacustrine and fluvial deposits. The Humber Falls Formation rests disconformably on top of the Rocky Brook Formation and has a maximum thickness of 250 m. It has been interpreted by Belt (1969) and Hyde (1979) to be an alluvial fan-fluvial deposit composed of grey to reddish conglomerate and arkosic sandstones and red to grey siltstones and mudstones. The Little Pond Brook Formation reaches a thickness of 750 m and is considered to be the fluvial (Hyde & Ware 1981) lateral equivalent of the Humber Falls Formation. It is composed of grey to red clastics and grey coarse-grained sandstone and pebble conglomerate interbedded with red fine-grained sandstone and siltstone. Spore assemblages in the Deer Lake Group suggest a Visean to possibly early Namurian age (Hyde 1984).

Two northeast-trending synclines separated by an intervening anticline deform the Deer Lake Group, the Humber syncline in the west and the Grand Lake syncline in the east. Bedding attitudes vary from sub-horizontal to overturned. It has been suggested by Hyde (1979) that folding of the Deer Lake Group was the result of dip-slip movement along major northeasterly trending faults. These faults controlled the overall form and sedimentation of the basin and may have been active, with strike-slip movement along them, during the deposition and deformation of the older Anguille Group.

3 SAMPLING AND METHODS

As Irving & Strong (1984) have pointed out, access is difficult and exposures are few and often weathered. Sampling was performed using the collection sites of Irving & Strong (1984) as guidance, however many sites are along stream beds and the water level was still too high for sampling during both field seasons of June and August of 2004. Nevertheless, 18 sites were sampled from all four units of the Deer Lake Group, mostly along lakeshores over a stratigraphic thickness of about 2000 m (Fig. 1). Many sites were not ideal: they were either too weathered or coarse-grained, which made drilling very difficult and resulted in many cores breaking. As a result, many short cores were collected, so sampling sites ranged from $\sim$8 to $\sim$15 cores that yielded one to three samples each. Cores were collected with a gasoline-powered 25-mm-diameter diamond coring drill and oriented using a magnetic compass.

The magnetic mineralogy of the Deer Lake group rocks was investigated by a variety of techniques. IRM acquisition experiments in fields up to 5 T applied with an ASC Scientific impulse magnetizer were used to identify the major coercivity components. IRM acquisition modelling was conducted (Kruiver *et al.* 2001). Thermal demagnetization of composite IRMs of 0.1, 0.3 and 1.2 T was performed using an ASC TD-48 thermal demagnetizer. These experiments were used to identify magnetic mineralogy (Lowrie 1990). Magnetic remanence was measured with a 2G Enterprises superconducting magnetometer at Lehigh University. Low-temperature magnetic properties were measured with a Quantum Designs MPMS cryogenic susceptometer at the Institute for Rock Magnetism, University of Minnesota. Specimens were cooled down to 20 K in zero

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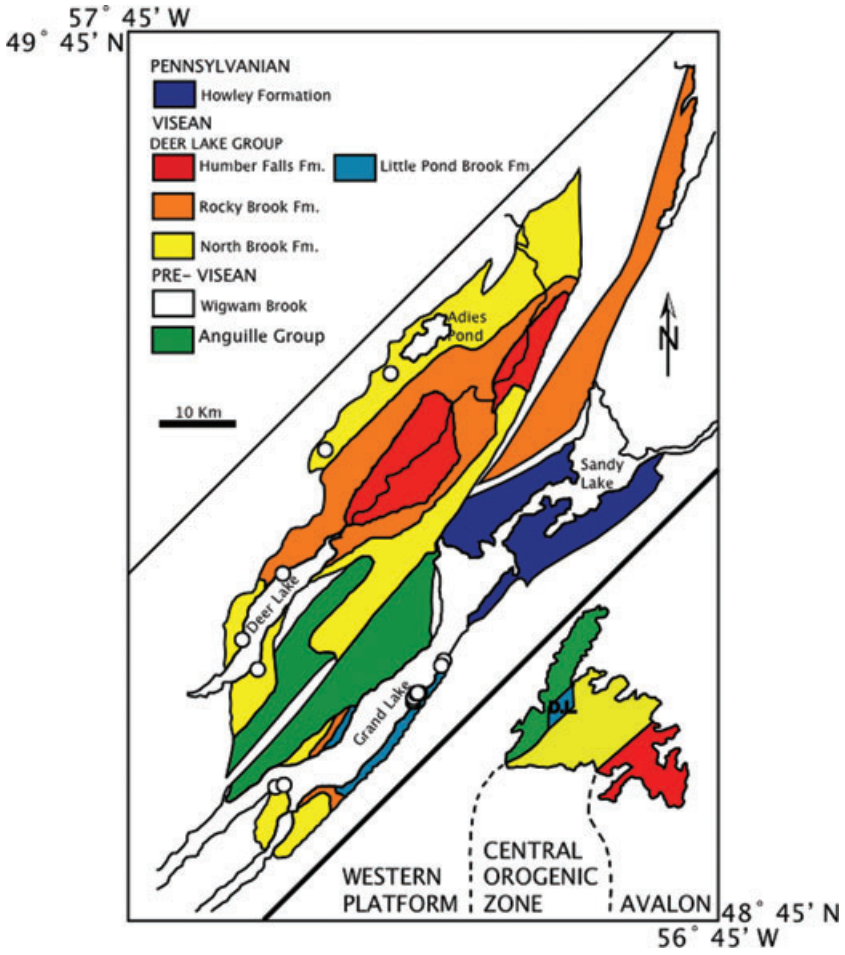


Figure 1. Location map of the Deer Lake Group of Newfoundland. Sampling sites are shown as circles on the map modified after Irving & Strong (1984). For a colour version of the figure, the reader is referred to the online version of this paper.

field and the remanence was measured on heating to 300 K and on cooling back to 20 K. First-order reversal curves (FORCs) were measured with a variable temperature Princeton Measurements Corporation vibrating sample magnetometer at the Institute for Rock Magnetism and processed using Winklhofer & Zimanyi's (2006) MATLAB code.

Thermal demagnetization was conducted in the thermal demagnetizer using 16 steps between 200 and 685 °C to identify the characteristic remanence of the rocks. Chemical demagnetization was conducted by leaching the samples in 3N HCl using 4-day leaching steps up to a cumulative 12 days and then 7-day steps up to a cumulative 40 days. AF demagnetization was conducted on representative samples in steps of 5 mT up to 50 mT and then 10 mT steps up to 100 mT.

Determining the remanence anisotropy of the characteristic remanence-carrying grains involved several steps. First, the coercivity spectra of the ferromagnetic grains was determined from partial anhysteretic remanence magnetization (pARM) using a Schonstedt GSD-5 tumbling specimen demagnetizer modified to apply pARMs. pARM acquisition curves were acquired in 10 mT intervals between 0 and 100 mT to determine the median coercivity window. Second, the magnetic fabric of these grains was measured by anisotropy of anhysteretic remanence (AAR) in a 10–70 mT coercivity window, with a 0.1 mT dc field. AAR was determined following the method described by McCabe *et al.* (1985): pARMs were applied in nine different orientations separated by 100 mT alternating field demagnetization steps. The nine pARMs were used to determine the best-fit anisotropy tensor by least squares. Finally, the individual particle anisotropy, or *a* factor, was measured directly following several steps. First, the samples were crushed, ball milled and finally sieved to ensure that the rock particles are smaller than 4 Φ (62.5 μm). Then, a slurry was made with the resulting crushed rock and distilled water which was then placed in an ultrasonic cleaner for at least 24 hr before circulating it in a magnetic extraction apparatus (Hounslow & Maher 1999). The magnetic separate was collected over a period of 2–3 weeks. Small amounts of the magnetic separates were mixed in a slow-drying epoxy resin for 24 hr and placed in a dc magnetic field (45 and 55 mT) to align the grains. Because the grains are aligned, the bulk AAR anisotropy of the epoxy samples provides an average individual particle anisotropy.

ChRMs were corrected for inclination shallowing using the relationship between magnetite anisotropy principal axes and inclination error described by Jackson *et al.* (1991):

$$\tan(I_0) = \frac{q_x(a+2)-1}{q_z(a+2)-1}\tan(I_m), \quad (1)$$

where I_0 is the corrected paleomagnetic inclination; I_m is the measured paleomagnetic inclination; q_x and q_z are the normalized maximum and minimum axes of the magnetic anisotropy ellipsoid ($q_x + q_y + q_z = 1$); a is the individual particle anisotropy (the a factor = easy magnetic axis/hard magnetic axis). An inclination correction may use either AMS or AAR parameters (both anisotropy

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principal axes and particle anisotropy), because the magnitudes of the corresponding normalized principal axes are linearly correlated (Stephenson *et al.* 1986; Tan & Kodama 2002, 2003; Tan *et al.* 2003). Inclinations have been corrected using the measured *a* factor and the measured anisotropy principal axes.

4 RESULTS

4.1 Magnetic mineralogy

Thermal demagnetization of orthogonal IRMs of 0.1, 0.3 and 1.2 T (Lowrie 1990) reveals the presence of at least three components of magnetization (Fig. 2). The 1.2 T demagnetization curve shows a drop around 680 °C. Typically this drop is small as in samples NF11e2 and NF15e3, but for some samples, as in sample NF8e2 it is the most prominent feature. In some cases, as for example sample NF8e2, it also shows a distinct decrease of intensity between 350 and 450 °C. The intermediate (0.3 T) demagnetization curve has different behaviour in different samples: in some cases it shows a distinct drop at 580 °C after which it is mostly removed, as in sample NF11e2, but in others it persists up to 680 °C, as in samples NF8e2 and NF15e3, a slight decrease around 400–450 °C is also observed. The 0.1 T curve consistently shows a decrease at 350 °C and then drops sharply at 580 °C. Upon reapplication of IRMs in the same fields, but after complete thermal demagnetization, the intensities of all IRMs show a decrease with respect to their initial intensity.

IRM acquisition modelling of specimens NF15e-1 and NF8e-1 revealed three major coercivity components. The lowest component is acquired between 60 and 80 mT. An intermediate component peaks at 500 mT, whereas the highest component peaks around 2000 mT (Fig. 3).

The low-temperature measurements of specimen NF11j-1 clearly show an initial loss of intensity when heating from 20 to 50 K and then a second abrupt drop at around 119 K. The cooling curve only clearly shows the 119 K transition, although not as abruptly. Both transitions are best observed when plotting the derivatives of the two curves (Fig. 4).

Hysteresis loops were acquired in fields of 1.2 T to capture a broad spectrum of coercivities (Fig. 5a), however, the FORC diagram of specimen NF11j-2, that has representative behaviour, shows the magnetic mineralogy being dominated by one low (<100 mT) coercivity phase or group of phases (Fig. 5b).

4.2 Paleomagnetism

Thermal demagnetization of the Deer Lake Group rocks revealed different components of magnetization that are completely removed by demagnetization up to 600–625 °C. The lowest unblocking temperature component, denoted the A component, is completely removed by 200 °C, but it is not possible to fully resolve it with our demagnetization scheme (Fig. 6).

An intermediate unblocking temperature component of magnetization, denoted B, is sometimes observed between 200 and 300–400 °C (Fig. 6d). Although this component is not fully resolved and only defined by three demagnetization steps, it is southerly directed with positive inclinations (reverse polarity). The highest unblocking temperature component of magnetization, component C, is observed above 300–400 °C and is completely removed by 600–625 °C. This component has a northerly direction and negative inclination (normal polarity) (Figs 6b and d). A few samples, however, revealed an antipodal direction of component C that trends towards the origin starting at 200 °C and is completely removed by 600–625 °C. This component is always southerly directed and has a positive inclination (Figs 6a and e). Complete demagnetization up to 685 °C was performed on all samples, however no stable direction has been isolated above 600–625 °C (Figs 6c and f). The Deer Lake Group mean direction for this high unblocking temperature component of magnetization in stratigraphic coordinates is

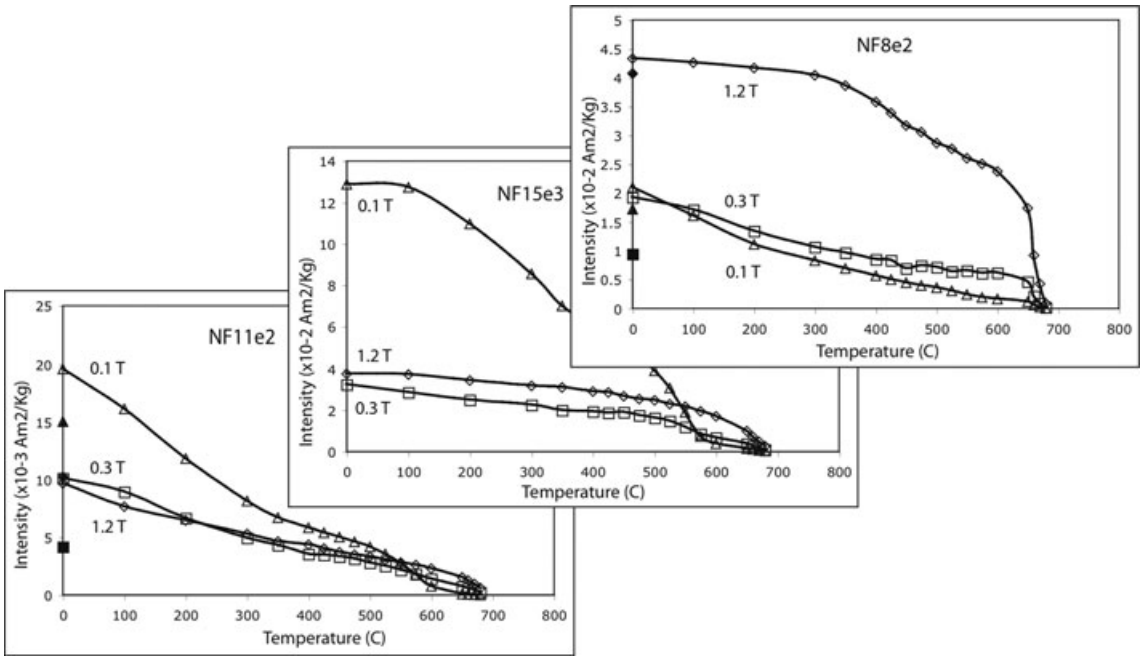


Figure 2. Thermal demagnetization of three orthogonal IRMs of 1.2 T (diamonds), 0.3 T (squares) and 0.1 T (triangles) (Lowrie 1990) for specimens NF8e2, NF11e2 and NF15e3. Filled symbols are re-applications of the IRMs at the end of the heating cycle (specimens NF8E2 and NF11E2). Loss of intensity on re-application of the IRMs indicates the presence of maghemite. Losses of intensity in the 350–450 °C range is indicative of maghemite whereas drops of intensity at 580 and 680 °C indicate magnetite and haematite, respectively.

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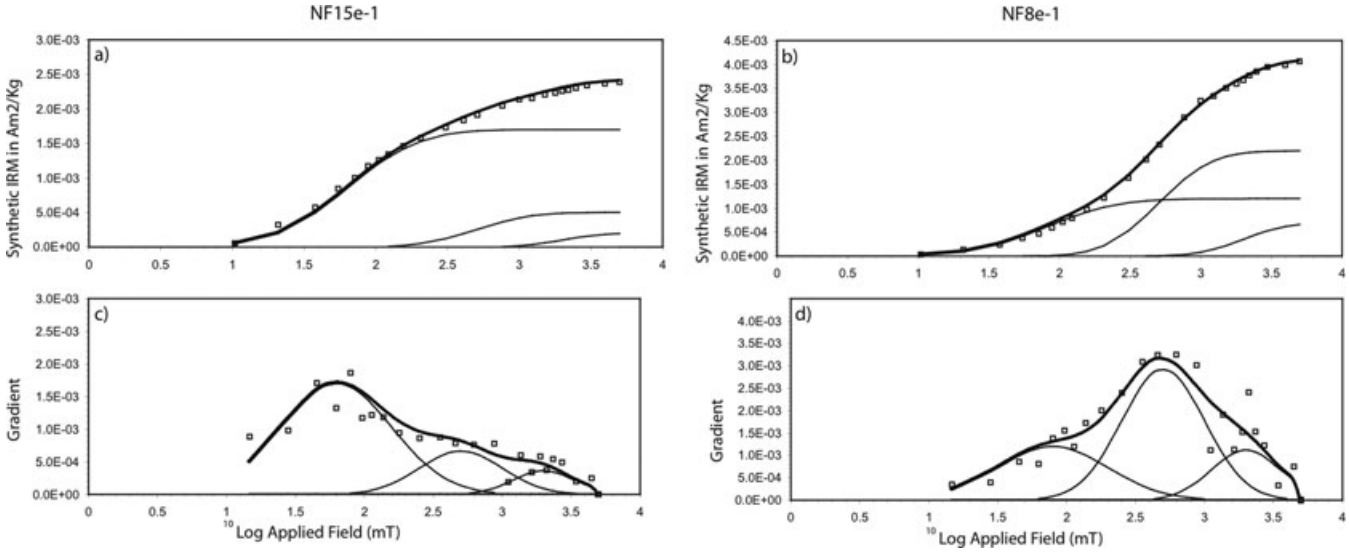


Figure 3. Modelling of IRM acquisition curves (Kruiver *et al.* 2001) for samples NF15e-1 and NF8e-1. The three peaks are in the coercivity ranges of magnetite (low), haematite (intermediate) and goethite (high).

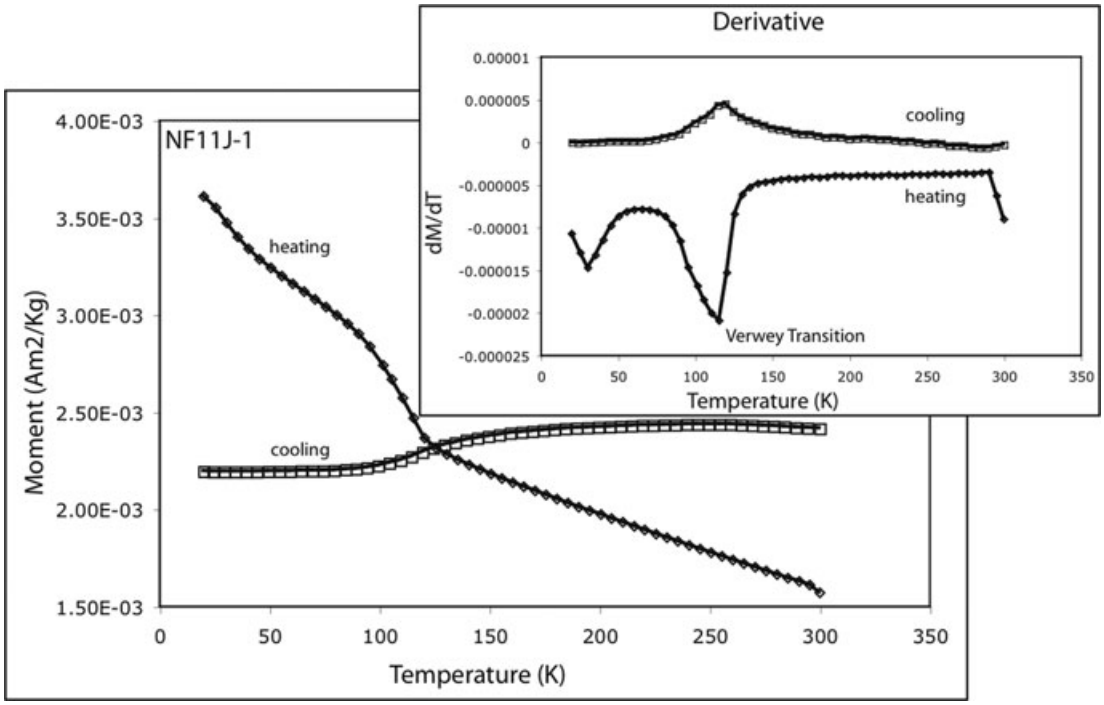


Figure 4. Low-temperature magnetization measurements for specimen NF11j-1 during heating from 20 to 300 K and cooling from 300 to 20 K all in zero field. The inset shows the derivatives of the two curves to highlight magnetization transitions. The initial loss of intensity at <50 K during heating is attributed to goethite. The Verwey transition, characteristic of magnetite is clearly visible around 119 K.

$D = 179.3^\circ$, $I = 30.5^\circ$, $\alpha_{95} = 19.7^\circ$, $N = 8$. Chemical demagnetization was not successful at isolating stable components of magnetization (Fig. 6g), so these results will not be treated.

For a few selected samples, AF demagnetization up to 50–60 mT isolated the same components of magnetization as thermal demagnetization (Fig. 6h). AF demagnetization up to 100 mT did not isolate a stable magnetization at higher fields (Fig. 6i). When the site mean directions from thermal demagnetization (Table 1) are plotted on a stereographic projection there is much scatter (Fig. 7a). Fold tests performed on these directions gave ambiguous results (Tauxe & Watson 1994), because the 95 per cent confidence interval for the fold test is extremely large, ranging from −58 to 151 per cent untilting. However, when the formation mean direction is plotted with its confidence ellipse together with the formation mean direction isolated by Irving & Strong (1984), the two directions are statistically indistinguishable (Fig. 7b). The directions isolated by Irving & Strong (1984) (Table 1) proved to be pre-folding directions of magnetization. We have combined our site mean directions with those of Irving & Strong (1984) and together, the site means pass the Tauxe & Watson (1994) and the McFadden & Jones (1981) fold tests, although with a large 95 per cent confidence interval (Fig. 8). The mean formation direction in stratigraphic

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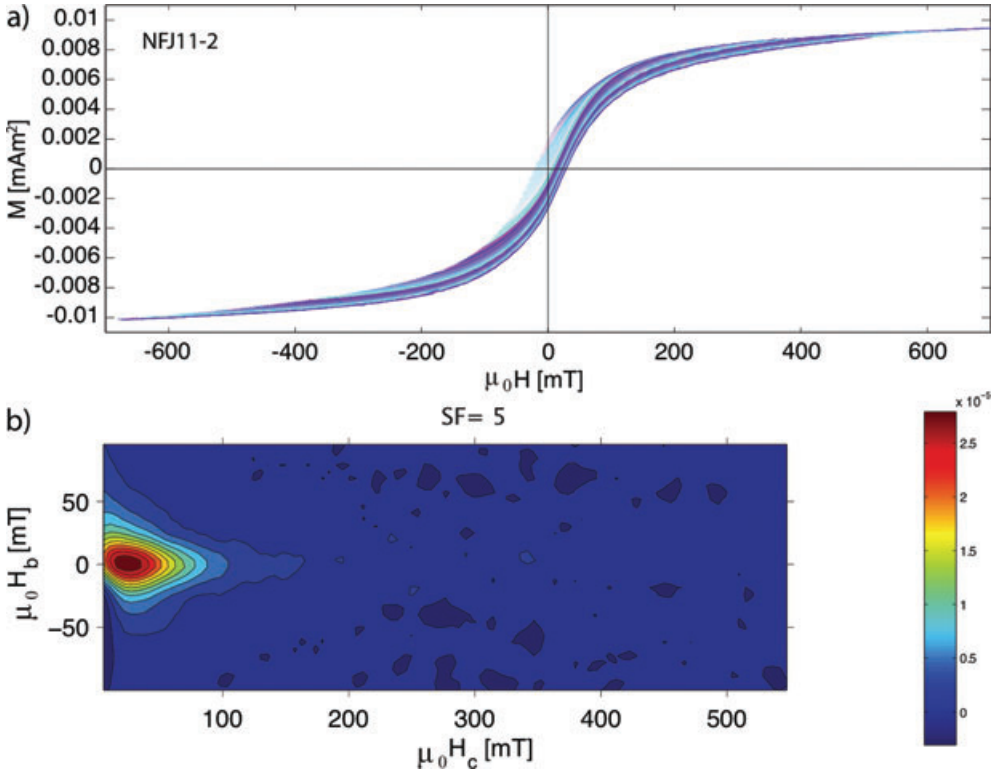


Figure 5. FORC diagram obtained for specimen NF11j-2 using the code of Winklhofer & Zimanyi (2006). This sample was chosen because samples from this site show typical thermal demagnetization behaviour. (a) Raw data and (b) FORC diagram (smoothing factor = 5) shows one main coercivity distribution that is <100 mT dominating the magnetization. For a colour version of the figure, the reader is referred to the online version of this article.

coordinates obtained by combining our data ($N = 8$) with that of Irving & Strong (1984) ($N = 21$) is $D = 179.7^\circ$, $I = 33.7^\circ$, $\alpha_{95} = 7.2^\circ$. This direction has been used for our inclination correction.

4.3 Magnetic Fabrics

AAR was used to measure the magnetic fabrics. To determine the coercivity range to apply the ARMs, we applied pARMs between 0 and 100 mT in intervals of 10 mT. The highest activation of magnetic grains, indicated by a peak in the pARM spectrum, occurred between 10 and 70 mT. We therefore measured the AAR in this coercivity range. Because the ChRM is demagnetized over the same AF interval (15–50/60 mT), the AAR is applied to the ChRM-carrying grains. The magnetic remanence fabric in stratigraphic coordinates (Fig. 9a) has a sub-bedding parallel foliation with very steep minimum axes plunging to the NW. The maximum and intermediate axes show good clustering at shallow inclinations. The maximum axes define a strong magnetic lineation that plunges gently to the SSW, whereas the intermediate axes plunge to the ESE. Orientation of the mean eigenvectors of the fabric are better appreciated after re-sampling the data using the bootstrap technique of Constable & Tauxe (1990) (Fig. 9b). Plotted on a Flinn diagram (Flinn 1962) (Fig. 9c) and as histograms of the eigenvalues (Fig. 9f), the fabric appears triaxial tending towards oblate.

In geographic coordinates (Fig. 9d, e), the clustering of all principal axes is less well defined. The minimum axes plunge towards the SW at intermediate angles, whereas the maximum and intermediate axes are scattered along a great circle oriented NW–SE and gently dipping to the NE.

4.4 Individual particle anisotropy (a factor)

Because Blow & Hamilton (1978) found a dependence of inclination shallowing on porosity, DRM research has focused on the attachment of magnetic particles to clays during DRM acquisition and the effect of the attachment on inclination shallowing (e.g. Scherbakov and Scherbakova 1983; Anson & Kodama 1987; Deamer and Kodama, 1990; Jackson *et al.* 1991; Mitra & Tauxe 2009). Flocs are thought to be formed by magnetic particles embedded in clay minerals, which are in turn arranged in a hierarchical, fractal nature (Mitra & Tauxe 2009 and references therein). However, the true structure and, most importantly, the orientation of individual magnetic particles within the flocs is unknown. This raises an important question when considering particle anisotropy for inclination shallowing because it carries the implication that the magnetic anisotropy of the flocs could be more important than that of the particles embedded in them. However, the misalignment of magnetic grain easy axes in a floc, as well as the misalignment of mean floc "easy axes" is measured as the overall preferred alignment of magnetic particles by the bulk rock anisotropy and the individual magnetic particle anisotropy, as measured by the *a* factor, is still the basic "quantum" of magnetic anisotropy in the rock.

Two samples of magnetic separates have been prepared, mixed in epoxy, and dried in dc magnetic fields of 45 and 55 mT. AAR of the sample dried in a 55 mT field had an anisotropy of 204 per cent (Rees 1966) and an RMS error of 3.4 per cent. The *a* factor measured from this fabric is 5.39. This high value is inconsistent with previous measurements of magnetite *a* factors (Kodama 1997; Tan & Kodama 1998; Kodama 2009) and may be due to clumping of the magnetic particles in the epoxy sample.

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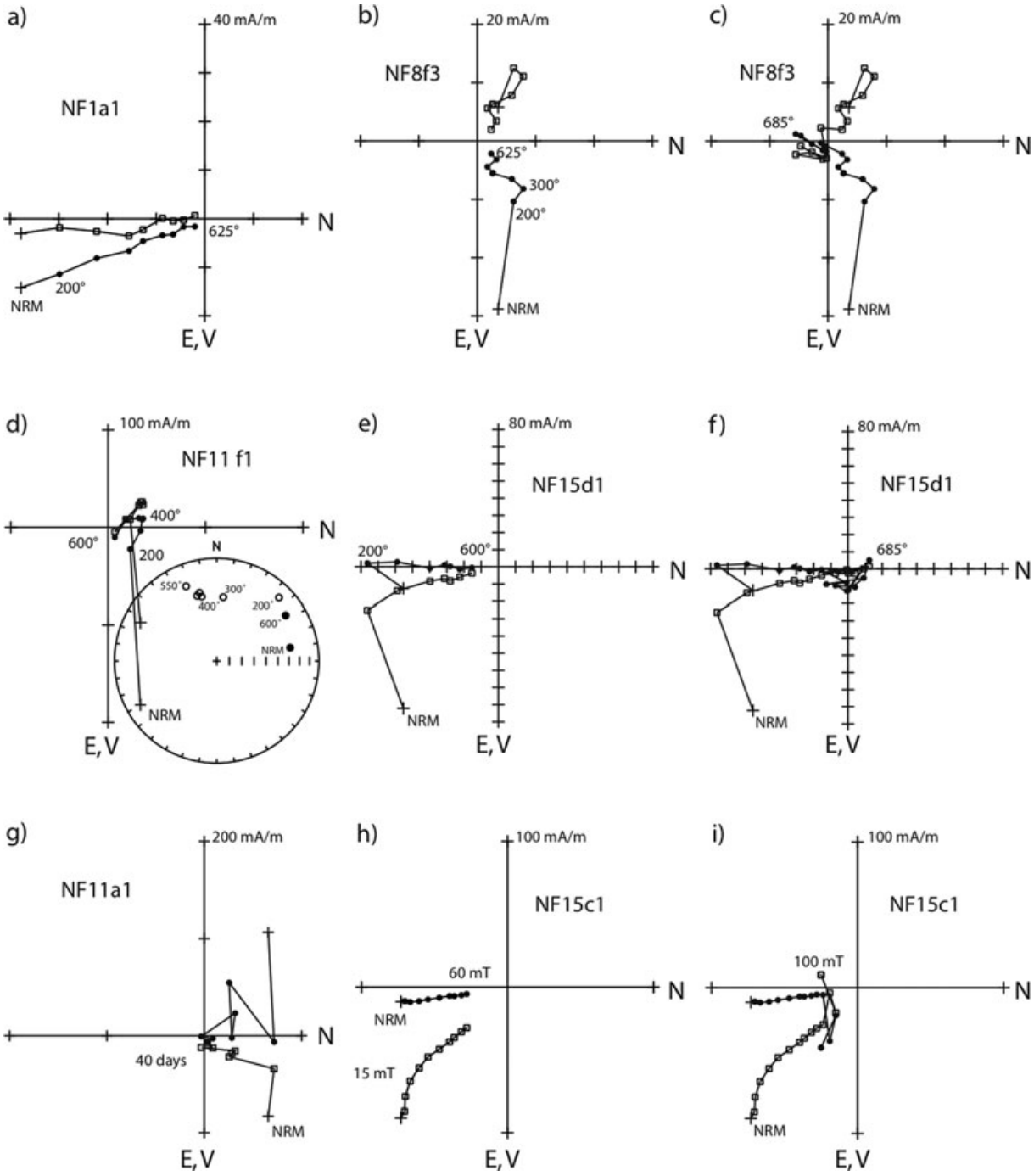


Figure 6. Orthogonal vector endpoint diagrams for thermal demagnetization (Zijderveld 1967). The ChRM was typically isolated between 200 and 600–625 °C. (a) and (e) show reverse polarity (south and down) components, whereas (b) and (d) show normal polarity (north and up) ChRM components; (g) shows the unstable behaviour of a chemically demagnetized sample, which is directly comparable to the behaviour observed in (d). The sample in (d) also reveals a poorly constrained lower unblocking temperature component that is antipodal to the ChRM and can be observed on the equal area projection between 200 and 400 °C, solid (open) circles are lower (upper) hemisphere projections. (h) Shows a result from AF demagnetization which is directly comparable to (e). To indicate that complete demagnetization was performed, but did not isolate a stable remanence above 600–625 °C or 50–60 mT, in (c) and (f) complete demagnetization is shown up to 685 °C of the same samples as in (b) and (e), where in (i) demagnetization up to 100 mT of the same sample as in (h) is shown. For all diagrams, solid (open) symbols are projections onto the horizontal (vertical) plane.

Confirmation of this interpretation is the observation that the ARM directions measured often diverged from the directions of the field used to impart the ARM, indicating strong magnetic interactions between the grains. We doubt the divergence of the imparted ARMs and the measured remanence is to be attributed to a gyroremanent magnetization effect (GRM) because this would result in remanent directions in an orthogonal plane to the ARMs imparted (Dunlop & Özdemir 1997), whereas the measured remanence for all the nine measurement positions used to calculate the anisotropy are subparallel to the orientation of the dc setting field. In contrast, the sample

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Table 1. Site mean directions isolated through thermal demagnetization. Sites nf1–15 are from this study, sites 06–85 are from Irving & Strong (1984) with the family of magnetization indicated, *in situ* declinations and inclinations are calculated from the reported directions in stratigraphic coordinates and bedding attitude, R and α_{95} (sites 22, 24, 67, 70, 71, 75) are calculated from the reported N and k.

Deer Lake Group ChRMs thermal demagnetization										
Site	*In situ D*	*In situ I*	Strike	Dip	Strat D	Strat I	*N*	*R*	*k*	α_{95}
nf1	160.7	5.9	347	30	158.6	1.9	5	4.9705	135.7	6.6
nf8	71.3	5.2	32	90	38.8	−39.1	5	4.8025	20.3	17.4
nf9	167.5	8.2	319	38	157.1	23.8	5	4.9024	41	12.1
nf10	191.6	21.5	327	39	169.7	44.1	5	4.9514	82.3	8.5
nf11	358.1	−9.7	310	35	347.5	−34	4	3.928	41.7	14.4
nf12	2.8	6.3	302	41	358.3	−29.1	4	3.4862	5.8	41.6
nf13	46.8	−13.7	285	23	52.5	−33	5	3.6377	2.9	54.4
nf15	170.8	18.9	39	5	169.3	17.7	4	3.9668	90.4	9.7
Mean	189	9.5	–	–	179.3	30.5	8	7.2111	8.9	19.7
06 S (N)	175	23.6	290	15	171	37	8	7.8542	48	8
09 S	166	15	210	30	177	34	8	7.6667	21	13
09 N	5.2	−13.5	210	30	195	24	1	–	–	–
09 N + S	168	14.8	210	30	179	33	9	8.6190	21	12
13 N	339.9	−32.4	150	20	171	27	1	–	–	–
22 S	198.3	8.3	280	27	200	35	2	1.9873	79	11.2
22 N	34.1	−9.7	280	27	219	34	2	1.9655	29	18.5
22 N + S	205.7	9.2	280	27	209	35	4	3.9167	36	16.0
24 S	170.7	29.2	225	30	189	51	2	1.9412	17	24.4
67 N (S)	65.8	−4.2	24	95	194	41	23	22.6857	70	4.0
70 N	4.9	−26.5	216	46	216	40	2	1.9091	11	30.6
71 S	156.6	28.2	170	25	171	31	1	–	–	–
71 N	326.9	−28.1	170	25	162	35	4	3.9667	90	10.0
71 N + S	330.8	−28.8	170	25	166	34	5	4.9437	71	9.0
75 S	159	26.3	303	26	144	39	3	2.9259	27	15.8
75 N	19.3	1.2	303	26	198	24	5	4.9111	45	12
75 N + S	185.9	9.3	303	26	181	32	8	7.1250	8	20
76 S (N)	171.7	14.2	280	22	165	28	6	5.7826	23	14
80 N	334.1	−34.8	53	11	153	24	1	–	–	–
82 S	146.7	26.8	45	8	146	19	1	–	–	–
85 S	178.3	−5.4	250	45	183	37	10	9.9692	292	3
Mean	181	16.9	–	–	179.7	33.7	29	27.1047	14.8	7.2

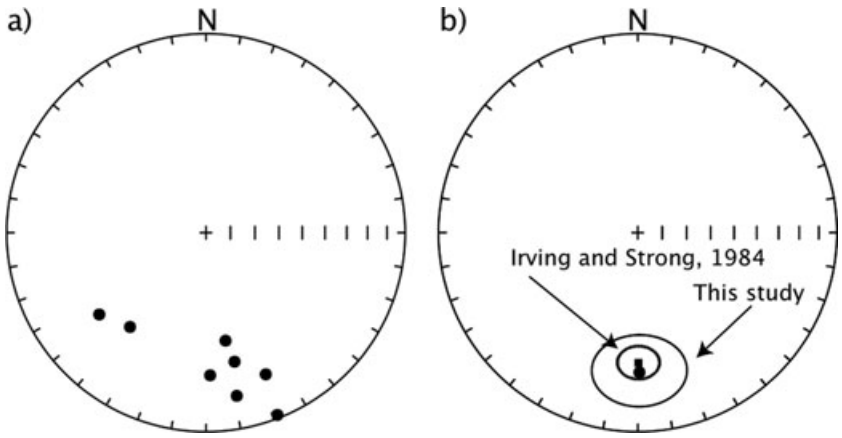


Figure 7. (a) Site mean directions in stratigraphic coordinates isolated through thermal demagnetization (all rotated to reverse polarity). (b) Comparison of means between the formation mean from this study (solid circle) and the formation mean from Irving & Strong (1984) (solid square).

dried in a 45 mT field shows an anisotropy of 84 per cent with an RMS error of 1.7 and an *a* factor of 1.99. This *a* factor value is consistent with previous observations for magnetite (e.g. Vaughn *et al.* 2005). In addition, ARM directions are closer to the applied field directions. The choice of a particle anisotropy of two was also confirmed by evaluating the symmetry of corrected site mean VGPs using the Bingham distribution (Bingham 1964) to analyse elliptically distributed data (Beck 1999). VGPs were re-sampled using the bootstrap technique of Constable & Tauxe (1990) to increase the sample number. Results from inclination corrections performed using a range of *a* factors (Bilardello 2008) indicate that the most symmetric and therefore geologically realistic distribution of site mean VGPs lies between *a* factors of 1.4 and 2.2.

Corrections performed using this range of *a* also agree, within error, to an inclination correction performed using the EI technique of Tauxe & Kent (2004) discussed earlier. This technique

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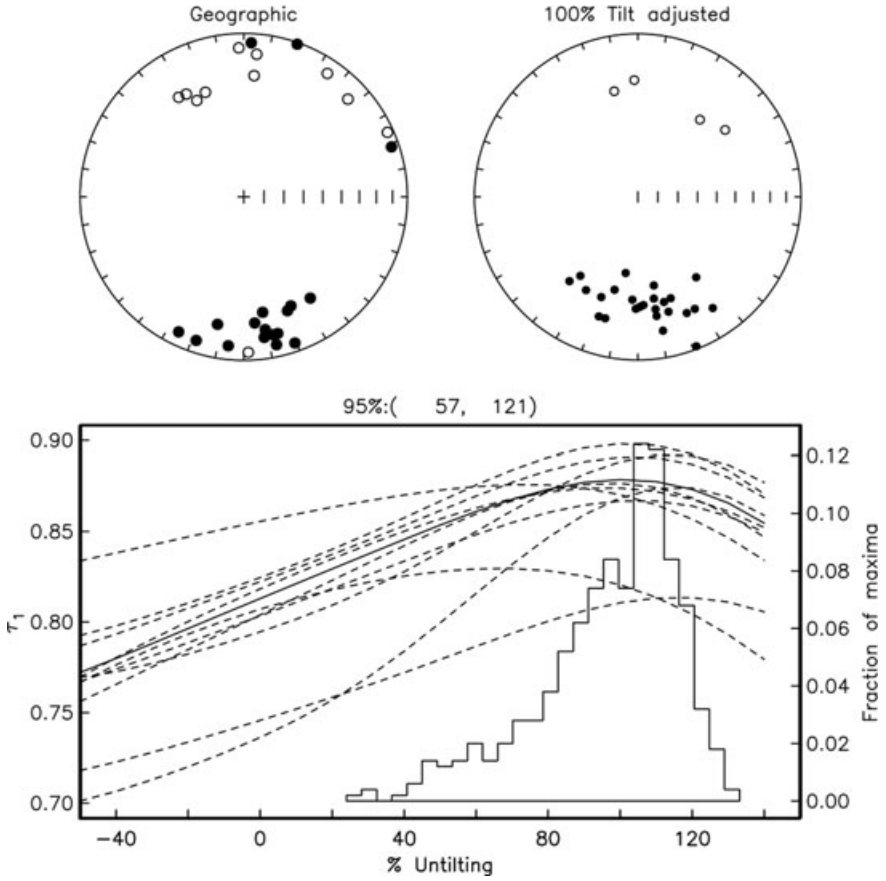


Figure 8. Fold test performed on data from this study and Irving & Strong (1984) combined.

converts symmetric VGPs to directions, which is essentially similar to evaluating the symmetry of corrected VGPs to determine particle anisotropy.

4.5 Inclination correction

The uncorrected paleopole of the Deer Lake Group lies at 22.2°N, 122.3°E, $A_{95} = 7.6°$. When an inclination correction is applied using an a factor of 2 and the measured AAR fabric, the mean direction of the Deer Lake Group is corrected to $D = 178.8°$, $I = 50.9°$, $\alpha_{95} = 6.3°$, $N = 29$, indicating an inclination shallowing of 17.2° (remanent inclination measured, $I = 33.7°$). When this direction is converted to a paleopole for the Deer Lake Group, its position plots at 8.4°N, 122.7°E, $A_{95} = 7.2°$, 13.8°S of the uncorrected paleopole.

An inclination correction was also performed using the EI technique of Tauxe & Kent (2004), which is based on the TK03.GAD model of the geomagnetic field using the software find EI (pmag-1.9.2 software package). This technique requires at least 100 sites to represent the distribution of directions drawn from plausible models of the geomagnetic field (Tauxe & Kent 2004). Because only 29 site mean directions were available, we used specimen directions obtained in this study together with the site mean directions reported by Irving & Strong (1984), which is justified because in sediments each specimen is a unique estimate of the geomagnetic field, giving 58 sample directions. The correction yields a new inclination of 41.1° (33° and 52°, low and high 95 per cent confidence interval inclinations), 9.8° smaller than the anisotropy-based correction, but with a large confidence interval that includes the previous result.

5 DISCUSSION

5.1 Magnetic Mineralogy

Results of the thermal demagnetization of three orthogonal IRMs (Lowrie 1990) suggest that haematite, magnetite and maghemite are present in the samples. The sharp drop observed at ∼680 °C is characteristic of haematite, however, this behaviour is not observed in the majority of our samples. All samples, however, show a loss of intensity, even if slight, at ∼580 °C, indicative of magnetite. The presence of maghemite is indicated by a slight loss of intensity between 350 and 450 °C (Dunlop & Özdemir 1997; McElhinny & McFadden 1999) and also so by the reapplication of an IRM at the end of the heating cycle, which shows an intensity much lower than before heating, assuming that the maghemite has inverted to haematite with heating. When the bulk susceptibility is plotted versus temperature (Fig. 10), a sudden increase is observed for certain samples starting around 450 °C, indicating that mineralogical changes are occurring upon heating. The increased susceptibility, coupled with the reduced IRM capacity suggests transformation to a phase with high IRM/χ (e.g. haematite) to one with very low IRM/χ (e.g. magnetite).

Modelling of IRM acquisition curves (Kruiver *et al.* 2001) shows coercivity peaks that correspond to the coercivities of magnetite/maghemite (50–70 mT) and haematite (500 mT), but also higher coercivity peaks (2000 mT), which we interpret to be goethite (Peters & Dekkers 2003) (Fig. 3).

MPMS low-temperature measurements confirm the results of the IRM modelling. The loss of intensity measured on heating between

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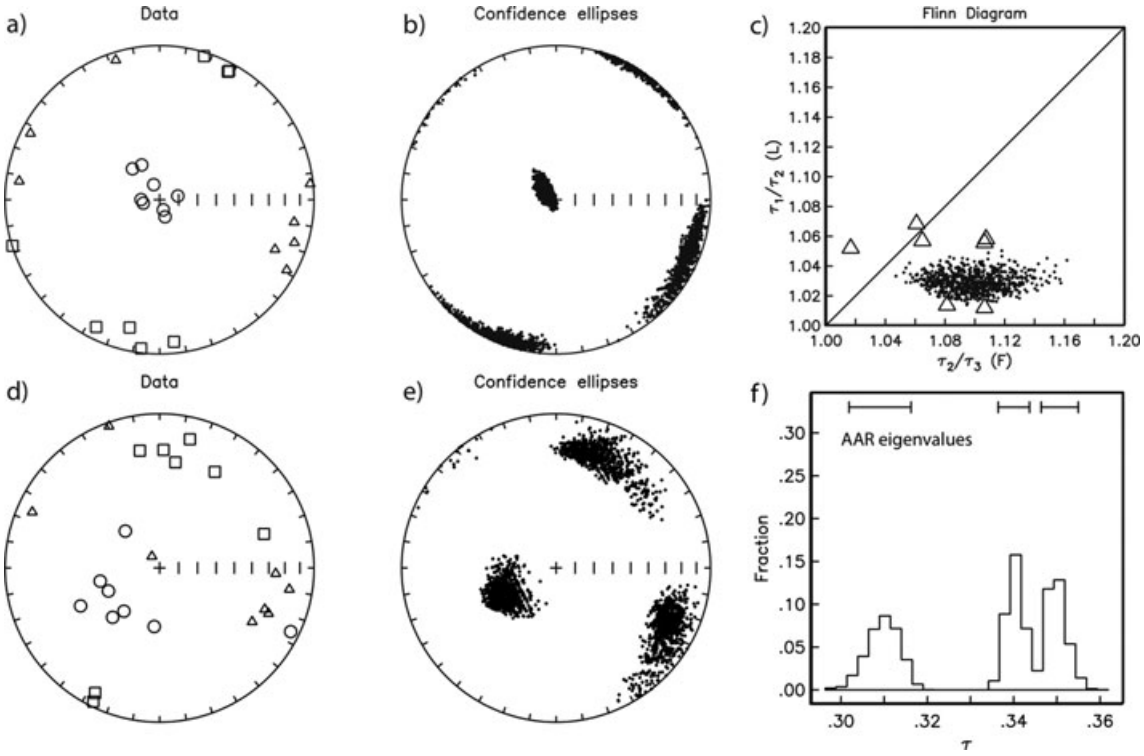


Figure 9. Magnetic fabric in stratigraphic coordinates measured with AAR. (a) original data; (b) bootstrap data; (c) Flinn diagram (Flinn 1962) showing a triaxial to oblate fabric. AAR magnetic fabric in geographic coordinates, (d) original data; (e) bootstrap data; (f) Histograms of the anisotropy eigevalues. Plots were generated using pmag-1.9.2 (Tauxe 2006).

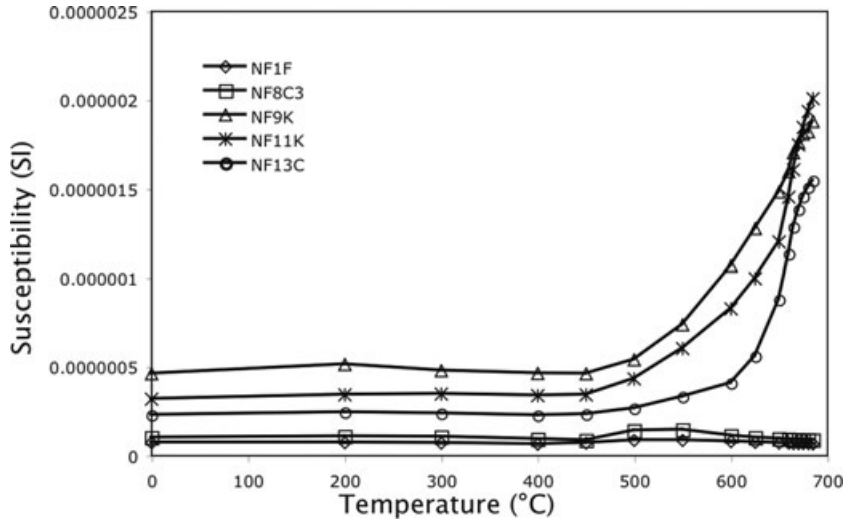


Figure 10. Bulk susceptibility as a function of temperature. Measurements where only possible for five sites for which we had spare samples. A sudden increase of susceptibily at about 450 °C was observed in three samples out of the five and is interpreted as an inversion of maghemite to haematite during heating.

20 and 50 K, although not diagnostic, could be attributed to goethite, as supported by the IRM acquisition modelling. Pyrrhotite also has a low-temperature transition at 34 K, however we believe the presence of sulphides to be unlikely in such highly oxidized rocks. The sharp drop at 119 K is consistent with the Verwey transition (Dunlop and Özdemir 1997) that occurs in magnetite at these temperatures. The small fraction of remanence lost on cooling through the transition is common for oxidized magnetite (Dunlop and Özdemir 1997). There is no indication of the Morin transition, diagnostic of haematite, around 250 K. For pigmentary nanophase haematite, the Morin transition is reduced and shifted towards temperatures as low as 5 K, until it is completely suppressed in particles ≤ 1 μm in size (Bando *et al.* 1965).

The FORC diagram, acquired in a peak field of 1.2 T, does not show the high-coercivity phases, but rather only one low-coercivity group below 100 mT. This is probably because the higher magnetization of magnetite completely covers the signal of haematite.

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5.2 Paleomagnetism

The characteristic remanent magnetization (ChRM) is typically revealed around 300 °C and is completely unblocked by 600–625 °C. Beyond these temperatures, no stable component of the remanence persists and it is not possible to isolate a direction.

AF demagnetization up to 50–60 mT isolated the same components of magnetization as the thermal demagnetization, indicating that the ChRM-carriers must be magnetic phases with low coercivities. Typically, these coercivities would be thermally demagnetized by 580 °C: magnetite has a Curie temperatures of 580 °C, whereas maghemite shows a loss of intensity in the 350–450 °C range that accompanies its inversion to haematite (Dunlop & Özdemir 1997; McElhinny & McFadden 1999). Demagnetization of 1.2 T IRMs, however, clearly reveals that haematite is present in the rocks, but its natural remanence is for the most part not stable as shown in the vector endpoint diagrams (Fig. 6). We therefore interpret the ChRM to be carried by magnetite and/or maghemite. The ChRM is only fully unblocked at temperatures of 600–625 °C. Thermally stable maghemite could, in theory, carry remanence up to its Curie point ∼645 °C (Dunlop & Özdemir 1997), alternatively, some finer grained haematite could also be contributing to the remanence.

The magnetization of the Deer Lake Group thus appears to be a pre-folding magnetization that has two antipodal, normal (north and negative) and reverse (south and positive), directions. In these red beds most of the ChRM is not carried by haematite of either detrital or chemical origin, but by magnetite/maghemite, with fine grained haematite only partly contributing to the natural remanence.

5.3 Magnetic fabrics

The fabric of magnetite/maghemite measured by AAR is sub-parallel to bedding and would indicate a primary depositional fabric. The fabric has its minimum axes steeply plunging to the NW. The maximum axes are clustered around the horizontal along a NNE–SSW lineation, whereas the intermediate axes show a clustering that gently plunges to the ESE. The magnetic lineation is sub-parallel to the regional scale fold axes, suggesting that the fabric could have been affected by strain. However, the beds that we have sampled do not follow the trend of those structures, but dip predominantly to the NE and would not therefore record the same strain event.

Excluding the tectonic fabric hypothesis, a depositional fabric explanation remains a plausible interpretation for the formation of the pronounced magnetic lineation. The minimum axes of the fabric, however, are very steep with the confidence ellipse including the pole to bedding and likewise, the maximum and intermediate axes are clustered on the horizontal, which means that the imbrication of the fabric is not statistically significant. This could be the result of burial compaction leading to a bedding parallel fabric. We interpret the fabric as a depositional fabric, with good clustering of the principal axes in stratigraphic coordinates, that has been affected by post-depositional compaction.

5.4 Inclination shallowing

Results of the inclination correction on the Deer Lake paleopole position are consistent with other inclination corrections performed so far on the Maringouin and Shepody formations of the Maritime Provinces of Canada (Bilardello 2008; Bilardello & Kodama 2010). The 17.2° of correction for the Deer Lake Group is larger in magnitude than the ∼10° of correction for the Shepody and Maringouin

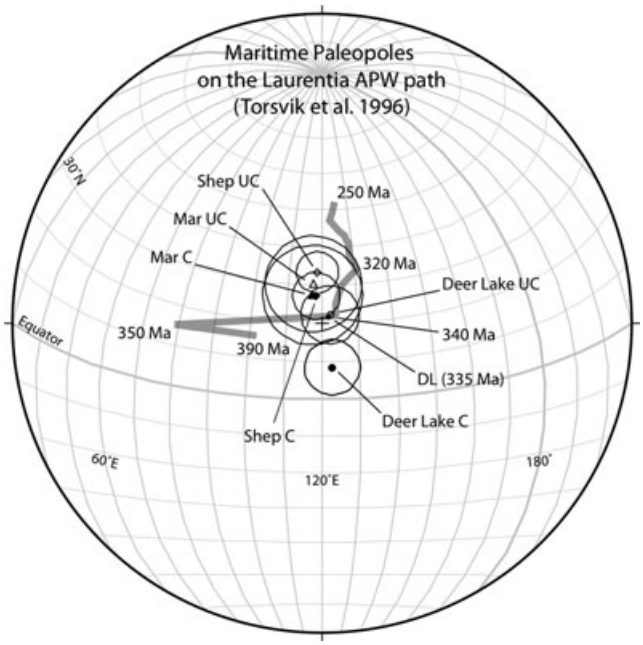


Figure 11. Uncorrected and corrected paleopoles from the Maritime Provinces of Canada. Large open (filled) circle, uncorrected (corrected) Deer Lake (UC and C) Group (this study); small filled circle, Deer Lake Group (DL) (Irving & Strong 1984); Open (filled) diamond, uncorrected (corrected) Shepody Fm. (Shep UC and C) (Bilardello 2008; Bilardello & Kodama 2010); Open (filled) triangle, uncorrected (corrected) Maringouin Fm. Paleopoles (Mar UC and C); Grey shaded line is the fitted APW path for Laurentia from Torsvik *et al.* (1996) rotated into North American coordinates.

Formations. Because the amount of inclination shallowing is dependent on the magnetic field inclination at deposition, which is in turn a function of latitude, it is not surprising that the Deer Lake Group, deposited at higher latitudes with steeper initial inclinations, has suffered from more inclination shallowing. Our results confirm the need to move the Lower Carboniferous portion of North American's APWP southward (Fig. 11) and strongly suggest a revision of the red bed-derived Paleozoic and Mesozoic portion of the APWP.

The southward translation of this portion of the APWP required by the corrections is ∼12° for the Middle Mississippian and ∼6° for the Upper Mississippian–Lower Pennsylvanian. As discussed in Bilardello (2008) and Bilardello & Kodama (2010), this magnitude translation would place Newfoundland in an arid climate zone during the Middle Mississippian, consistent with the evaporitic deposits present in the Nova Scotia and New Brunswick Carboniferous rocks (Geldsetzer *et al.* 1980). Newfoundland would then move progressively northwards towards more humid conditions in the Middle to Upper Pennsylvanian, as indicated by the thick coal deposits that characterize the Cumberland and Pictou groups (Hacquebard 1970; Hacquebard & Donaldson 1969).

One other important effect of the southward translation of North America (see Bilardello & Kodama 2010) is to increase the overlap between North America and Gondwana in the early Pangea assemblages (Van der Voo 1993). Our results therefore favour a Pangea B assemblage (Muttoni *et al.* 2003, 2009), in which North America is positioned to the west of Gondwana, thus eliminating the overlap.

In Bilardello & Kodama (2010), however, we argue that to fully evaluate the effects of the correction it is necessary to consider the placement of Gondwana for that time period. This is because

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the Carboniferous Gondwana paleopoles are derived primarily from sedimentary rocks (Torsvik & Van der Voo 2002), which are likely to have been affected by inclination shallowing. If this were the case an inclination correction would move Gondwana further south, possibly eliminating the overlap.

5.5 Comparison with other inclination correction techniques

King (1955) suggested a relationship between the measured (remanent) inclination and field inclination to explain inclination shallowing observed during redeposition experiments. This relationship follows the equation:

$$\tan(I_m) = f \tan(I_0), \quad (2)$$

where I_m is the measured, or remanent inclination; I_0 is the field inclination and f is the shallowing factor that has been observed to lie between 0.4 and 0.55 from experimental data (King 1955; Tauxe & Kent 1984).

We have collected inclination shallowing data for both magnetite and haematite from various sources and compiled it with our data from the Maritime Provinces of Canada (Table 2). All the results for magnetite are from anisotropy-based inclination shallowing corrections, enabling direct calculation of the f factors, which range from 0.54 (Deer Lake Group) to 0.79 (Nacimiento Fm).

For haematite, Tauxe (2005) and Kent & Tauxe (2005) have studied Oligo-Miocene and mid Miocene haematite-bearing red sedimentary rocks from Asia and Pakistan, and Late Triassic red beds from around the North Atlantic Ocean, respectively and have applied an inclination shallowing correction using a statistical geomagnetic field model (TK03.GAD), which relies on the elongation distribution of paleomagnetic directions caused by secular variation (Tauxe & Kent 2004). Their technique yielded f factors from both studies between 0.4 and 0.77. Our results from haematite-bearing rocks of the Maritime Provinces of Canada imply f factors that range from 0.64 to 0.83. The result of Tan *et al.* (2003) from Chinese Cretaceous red beds was also included.

We have deliberately left out values obtained from redeposition experiments because when the redeposited sediments are measured in the zero field of the magnetometer the remanent inclination may be affected by grains physically reorienting in the high water content sediments and therefore the measurement would not represent a true DRM (Tauxe & Kent 1984; Carter-Stiglitz *et al.* 2006). Likewise, results of Tan *et al.* (2007) from the Newark Basin red beds have not been included because Kent & Tauxe (2005) already include results from the same Newark Basin Formations in their study.

Even higher inclination errors have been observed for Neogene red sediments from the Vallès-Penedès basin (Garcés *et al.* 1996). Site mean inclinations of less than 20° are reported for three out of five finely laminated red bed sites, whereas they report a gamut of inclinations ranging between 32° and 56° from six more sites collected from finely laminated, burrowed, edafized and massive facies. The expected geomagnetic field inclination is 60°. Garcés *et al.* (1996) never report statistics at the Formation level, however, calculation of a Formation mean (this study) from the 11 reported site means yielded $D = 0.6°$, $I = 34.7°$ and $\alpha_{95} = 10.2°$.

Magnetite-bearing rocks display a range of f values between 0.54 and 0.79 with a mean value of 0.65 and a standard deviation of 0.08 ($N = 9$). Haematite-bearing rocks have values between 0.4 and 0.83 with a mean of 0.59 and a standard deviation of 0.12 ($N = 15$). These values correspond to greatly varying amounts of inclination shallowing for both magnetite- and haematite-bearing rocks (Fig. 12). It is very tempting to take the mean values and their standard errors (0.026 for magnetite and 0.031 for haematite) to perform simplified inclination corrections of formations that are suspected to have suffered from inclination shallowing. However,

Table 2. Measured inclinations (I_m), corrected inclinations (I_c) and derived f factors for inclination corrected magnetite- and haematite-bearing sedimentary rocks.

		Magnetite and haematite f factors			
Formation/group	Magnetic carrier	I_m	I_c	f	Reference
Glenshaw Fm.	Magnetite	23.4	33.7	0.65	Kodama (2009)
Perforada Fm.	Magnetite	35.6	50.1	0.60	Vaugn *et al.* (2005)
Nanaimo Group	Magnetite	50.9	60.5	0.70	Kim & Kodama (2004)
Valle Group	Magnetite	19.1	26.6	0.69	Li *et al.* (2004)
Ladd Fm.	Magnetite	46	58	0.65	Tan & Kodama (1998)
Point Loma Fm.	Magnetite	39.5	56	0.56	Tan & Kodama (1998)
Pigeon Point Fm.	Magnetite	41.6	51.5	0.71	Kodama & Davi (1995)
Deer Lake Group	Magnetite	33.7	50.9	0.54	This study
Nacimiento Fm.	Magnetite	51.3	57.7	0.79	Kodama (1997)
Kapusaliang Fm.	Haematite	29	52.5	0.43	Tan et al. (2003)
Shepody Fm.	Haematite	20.4	30.1	0.64	Bilardello & Kodama (2010)
Maringouin Fm.	Haematite	24.9	29.1	0.83	Bilardello & Kodama (2010)
Dan River	Haematite	5.9	10	0.59	Kent & Tauxe (2005)
Newark basin: Princeton	Haematite	5.2	9	0.57	Kent & Tauxe (2005)
Newark basin: Nursery	Haematite	8.8	21	0.40	Kent & Tauxe (2005)
Newark basin: Titusville	Haematite	13	20	0.63	Kent & Tauxe (2005)
Newark basin: Rutgers	Haematite	14.2	21	0.66	Kent & Tauxe (2005)
Newark basin: Somerset	Haematite	15.7	24	0.63	Kent & Tauxe (2005)
Newark basin: Weston	Haematite	17.5	33	0.49	Kent & Tauxe (2005)
Newark basin: Martinsville	Haematite	18.2	34	0.49	Kent & Tauxe (2005)
St. Audrie's Bay	Haematite	33.4	44	0.68	Kent & Tauxe (2005)
Jameson Land	Haematite	45.1	60	0.58	Kent & Tauxe (2005)
Subei area	Haematite	43.7	63	0.49	Tauxe (2005)
Potwar Plateau	Haematite	33.7	41	0.77	Tauxe (2005)

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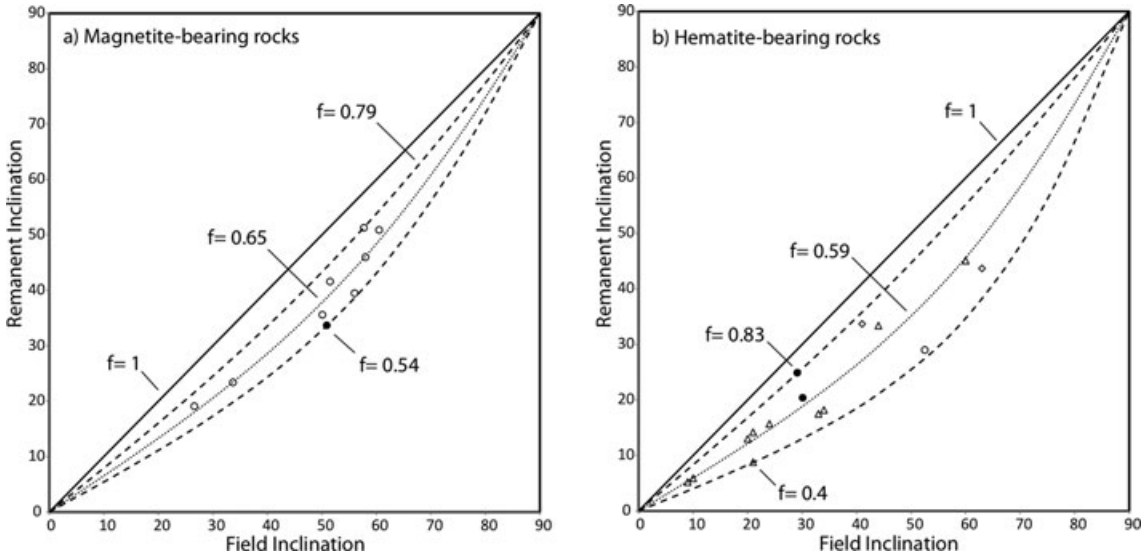


Figure 12. Plot of the remanent (measured) inclinations (I_m) as a function of the field (corrected) inclinations (I_0). (a) Magnetite-bearing sedimentary rocks, (b) is for haematite-bearing sedimentary rocks. The solid lines are the lines of perfect correlation between I_m and I_0 ($f = 1$), the dashed lines comprise the smallest and largest f factors measured, thus defining the ranges of maximum and minimum inclination shallowing observed. The dotted lines have been plotted using the mean f factors. (a) The solid circle is from this study; the open circles are from other magnetic anisotropy-based studies (see Table 2). (b) The solid circles are from our data (Bilardello 2008; Bilardello & Kodama 2010); the open circle in from the anisotropy-based correction of the Kapusaliang Fm.; open triangles and diamonds are from the TK30.GAD corrections of Kent and Tauxe (2005) and Tauxe (2005) , respectively (see Table 2).

the f factor, from the inclination correction formula, incorporates measurements of particle anisotropy and the magnetic fabric, which make this parameter non-unique. Although particle anisotropy may not vary substantially for haematite because it is not dependent on crystal shape (Dunlop & Özdemir 1997; Bilardello 2008; Bilardello & Kodama 2009), it is for magnetite (Dunlop & Özdemir 1997). Likewise the magnetic fabric of the rocks is dependent on the degree of alignment of the particles, which is expected to vary with magnetic mineralogy and process (Jackson 1991). The range of measured f factors is therefore more significant than the mean value, when evaluating inclination errors.

The maximum inclination error that can be expected from the range of compiled f factors and the tan–tan relation of King (1955) is of 17.4° for a magnetite-bearing rock with an original inclination of 55°, using the smallest observed f factor of 0.54 (Fig. 12), and of 25.3° for a haematite-bearing rock with an original inclination of 60°, using the smallest observed f factor of 0.4 (Fig. 12).

Inclination shallowing results may also be evaluated in terms of the dipole equation, $\tan I = 2\tan \lambda$, where λ is the latitude. When evaluating I_m versus the latitude calculated from the corrected inclinations, λ', the dipole relationship becomes

$$\tan(I_m) = \Lambda \tan(\lambda'), \quad (3)$$

where Λ is a proportionality constant, the 'paleolatitude correction' constant. Because the field inclination versus remanent inclination equation and the dipole equation have the same tan–tan form and are closely related, it is possible to calculate a corrected latitudinal range using $\Lambda = 2f$ for sedimentary rocks whose measured remanent inclinations are suspected to be shallow.

To test these relationships, we have applied simplified inclination corrections to our calculated mean ($D = 0.6°$, $I = 34.7°$, $\alpha_{95} = 10.2°$, $N = 11$) of the Vallès-Penedès basin red beds from the data of Garcés *et al.* (1996) (expected field inclination = 60°).

When we apply a correction using the observed haematite shallowing factor end members of 0.4 and 0.83, the corrected inclinations are 60° and 39.8°. A correction applied using the mean f factor of 0.59 yields an inclination 49.6°. This simple test indicates that for these red beds the smallest of the observed f factors restores the inclination to that of the ambient geomagnetic field whereas the mean or largest f factors observed would underestimate the correction by ∼10° and ∼20°, respectively. Nevertheless, this test shows what ranges of inclination corrections may be expected for red beds that are suspected to be shallow.

6 CONCLUSIONS

The inclination shallowing correction performed on the magnetite-bearing Deer Lake Group rocks using the measured AAR fabric and the measured particle anisotropy of the ChRM- carrying grains indicates 17.2° of shallowing. This amount of inclination shallowing corresponds to a 13.8° increase in colatitude for the corrected paleopole, a larger value than what has been obtained from other corrected red bed palepoles from the Maritime Provinces of Canada, but consistent with the tan–tan variation of measured, shallowed inclination as a function of field inclination. Together with the other inclination shallowing corrections that have been applied to this portion of the Paleozoic and Mesozoic APWP for North America, these results strongly suggest a revision of sedimentary rock, and in particular red bed- derived, APWPs. The corrected paleolatitude puts Newfoundland in an arid climate zone for the Lower Carboniferous, in agreement with lithologic indicators. The corrected paleogeography increases the overlap between North and South America in a Pangea A configuration, making a Pangea B reconstruction necessary for this time period.

However, to fully evaluate the effects of the correction it is necessary to consider the positioning of Gondwana.

The distribution of the corrected paleomagnetic directions provides a means of estimating the a factor used for the anisotropy correction, and results suggest that the corrected inclinations correspond to a geologically reasonable distribution of directions.

A comparison of the proportionality constant between the tangent of the remanent inclinations and the tangent of the field

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inclinations, the flattening factor f, derived from our anisotropy-based inclination corrections and f factors resulting from other studies show strong agreement. A compilation of f factors from inclination corrections performed on both magnetite- and haematite-bearing rock formations allowed to determine the range of f factors that may be expected, allowing determination of the possible ranges of inclination shallowing.

The Neogene Vallès-Penedès basin red beds have been corrected to test the range of derived f factors. The smallest of the f factors perfectly restores the inclination to that of the ambient field, whereas the mean and largest f factors considerably undercorrect the inclination. This simple test indicates what ranges of inclination corrections may be expected for sedimentary rocks suspected to be shallow.

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